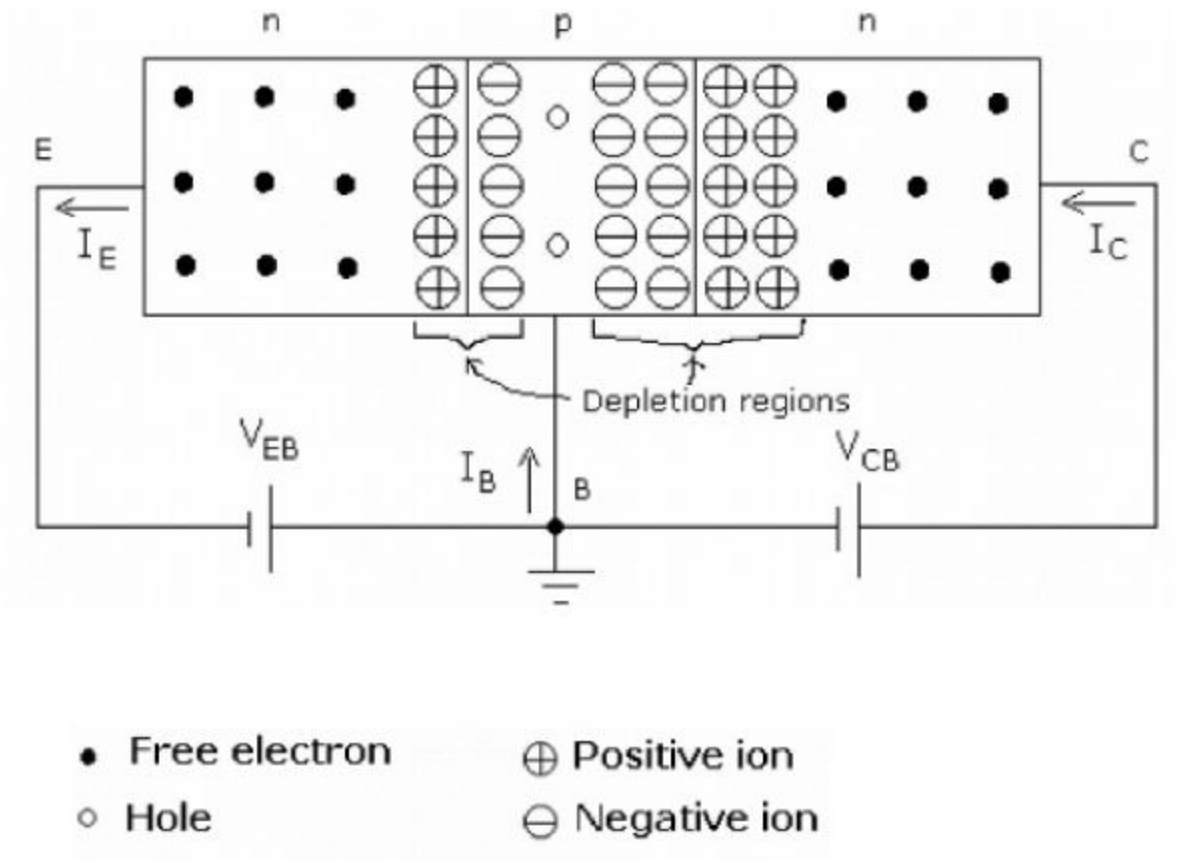
2. Bi-polar Junction Transistor

Modes of Operation

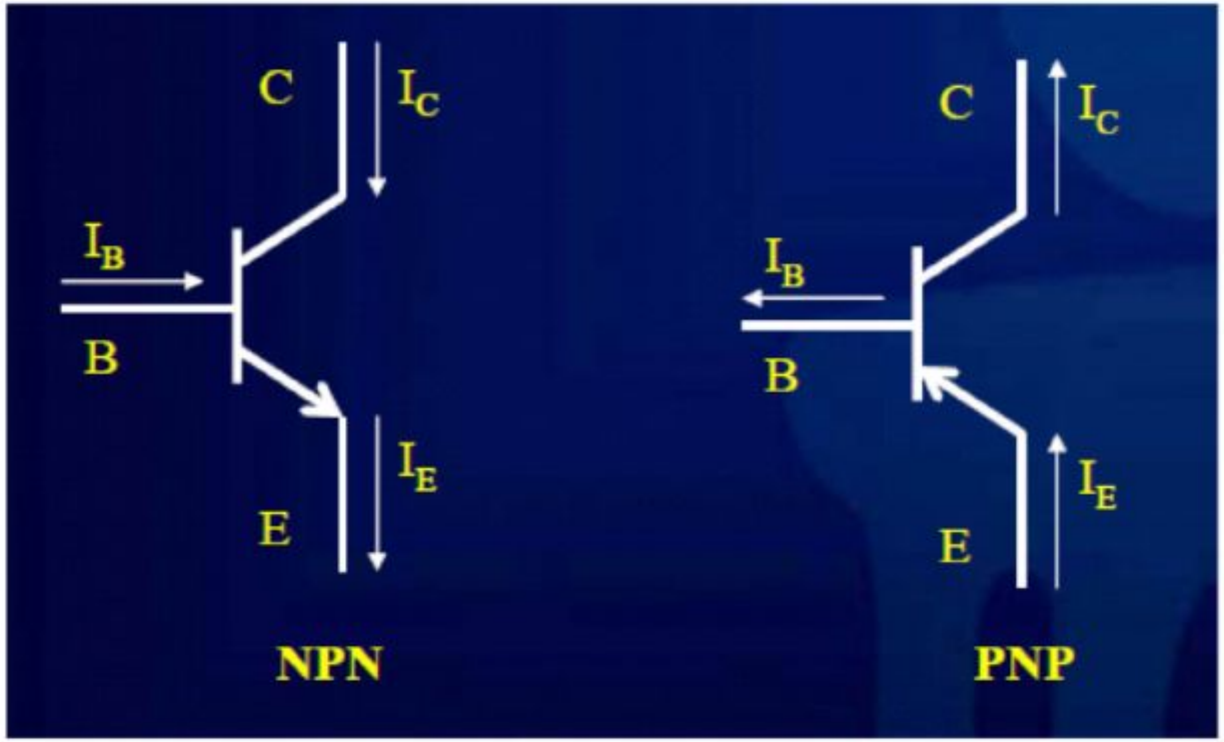
Modes	EBJ	CBJ	Application
Cutoff	Reverse	Reverse	Switching application in digital circuits
Saturation	Forward	Forward	
Active	Forward	Reverse	Amplifier
Reverse active	Reverse	Forward	Performance degradation

Transistor- Normal Operation



- When EB junction is forward biased, free electrons from emitter region drift towards base region
- Some free electrons combine with holes in the base to form small base current
- Inside the base region (p-type), free electrons are minority carriers. So most of the free electrons are swept away into the collector region due to reverse biased CB junction
- Three currents can be identified in BJT
  1. Emitter current;
  2. Base current;
  3. Collector current

Current directions in NPN and PNP transistors:

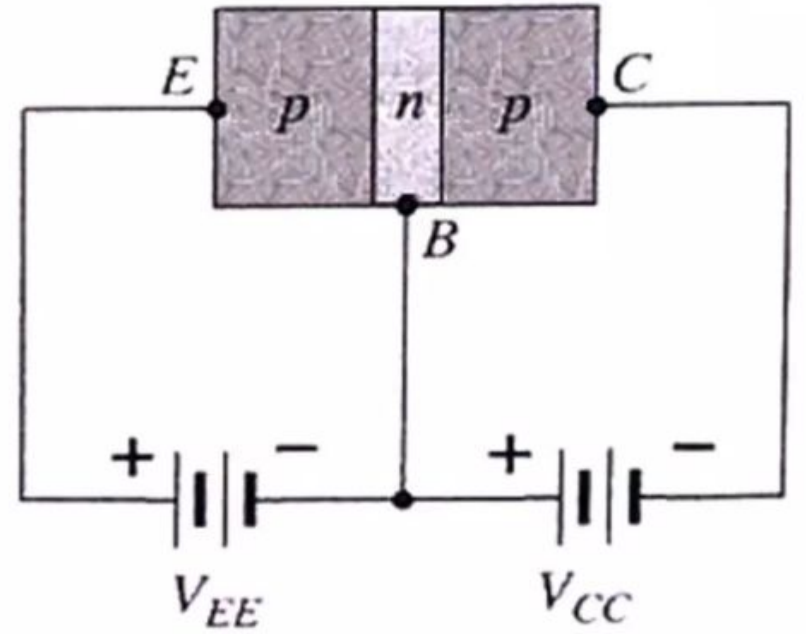


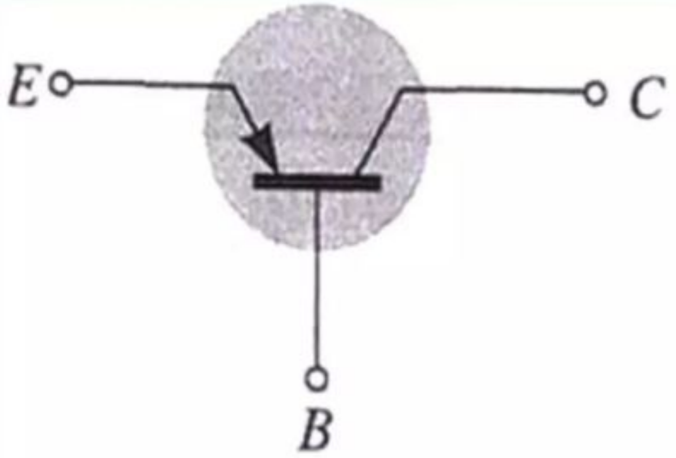
For both varieties: $I_E = I_C + I_B$ -----------(1)

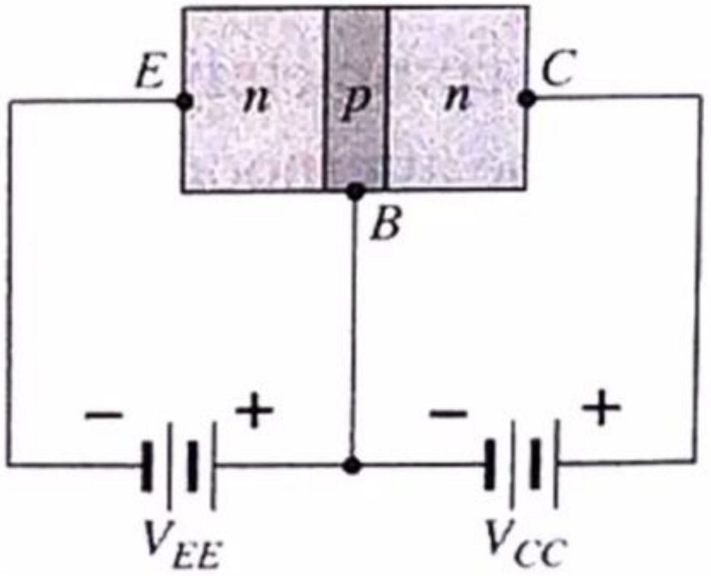
Transistor Configurations

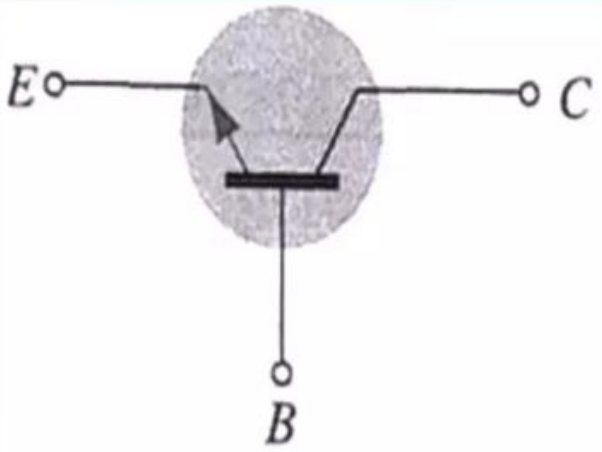
- Common-Base Configuration
- Common-Emitter Configuration
- Common-Collector Configuration

Common-Base Configuration

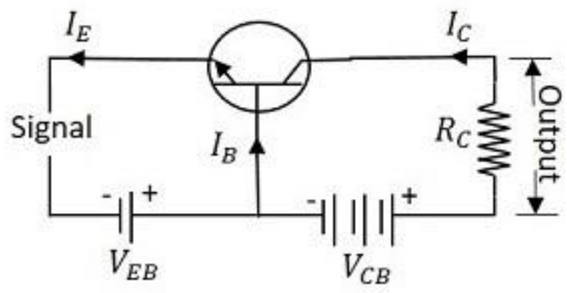




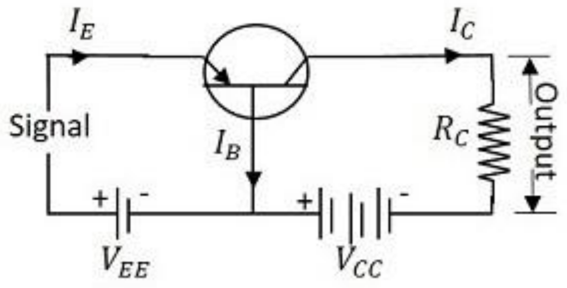




Common Base Configuration

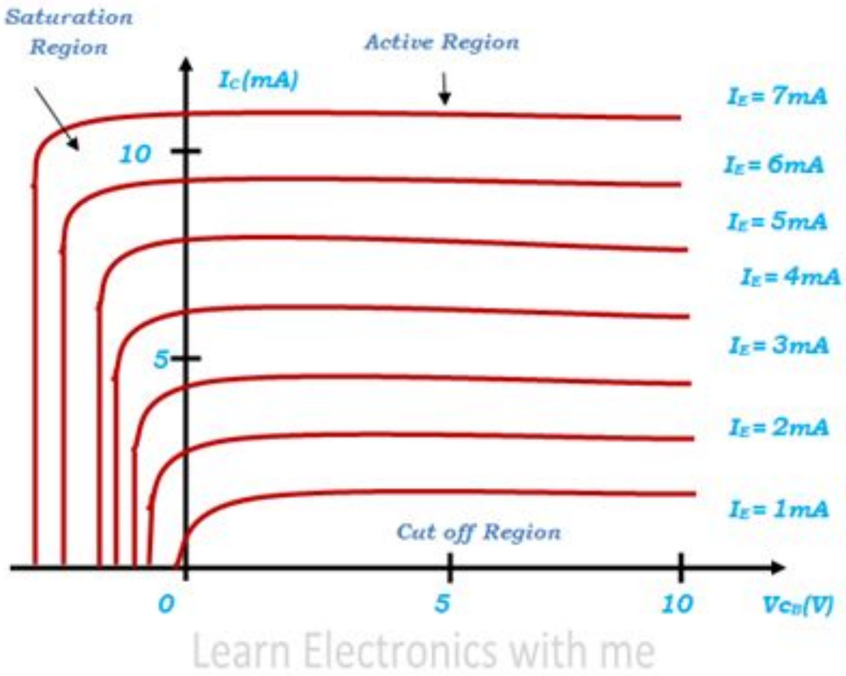


Using NPN transistor

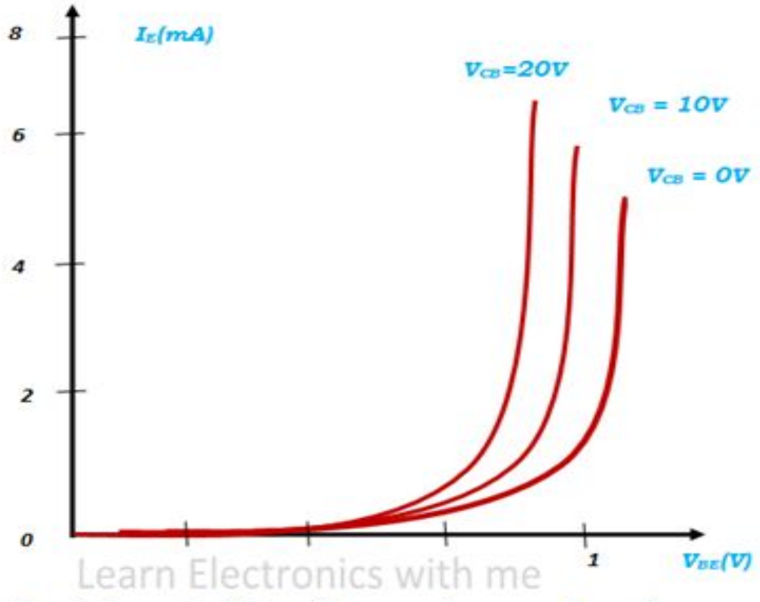


Using PNP transistor

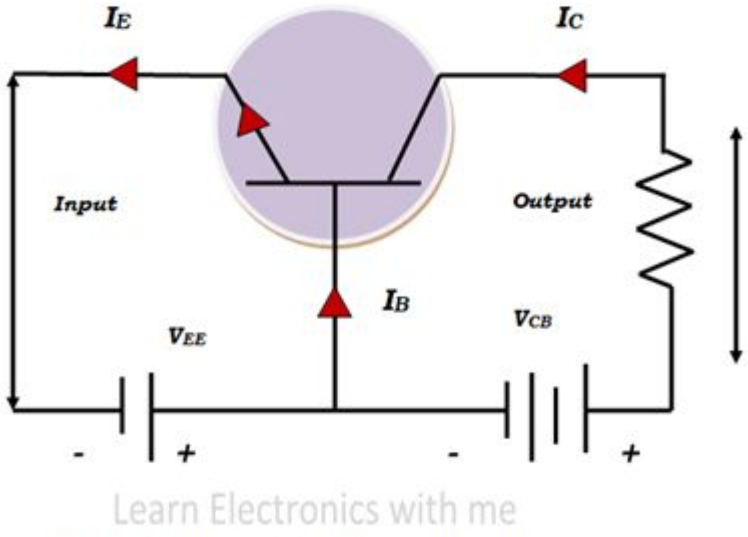
Common Base Configuration



Output characteristics of common base configuration

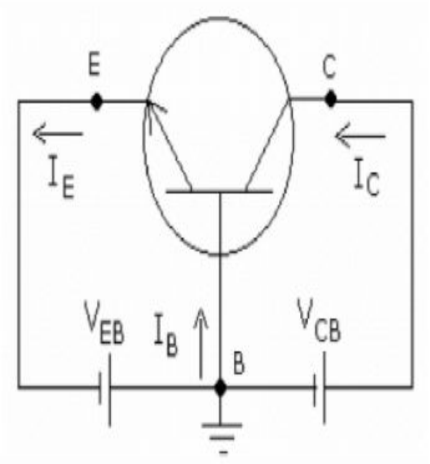


Input characteristics of common base configuration



Common Base configuration of NPN transistor

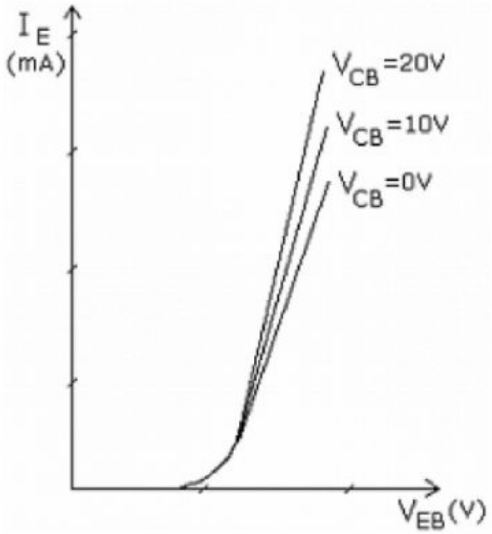
Common Base Configuration



(Resistors are not shown here for simplicity)

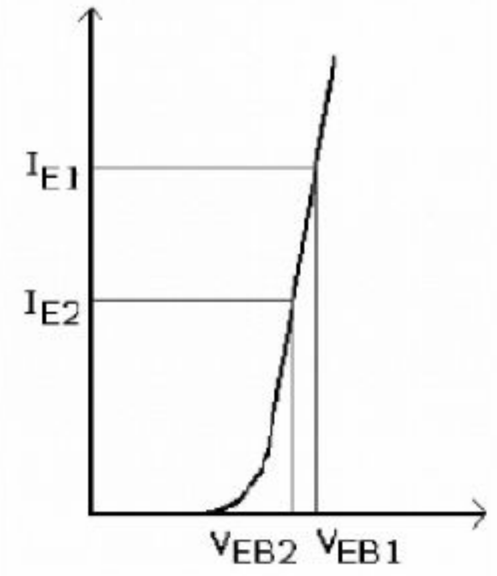
CB Input characteristics

– A plot of *IE versus VEB* for various values of *VCB*
– It is similar to forward biased diode characteristics
– As *VCB is increased, IE* increases only slightly
– Note that second letter in the suffix is *B (for base)*



Input resistance r_i

$$r_i = \left.\frac{\Delta V_{EB}}{\Delta I_E}\right| with\ V_{CB}\ const$$

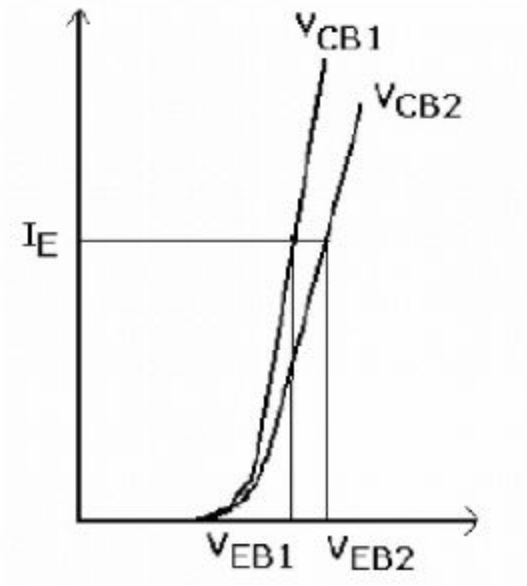


$$r_i = \left|\frac{V_{EB1} - V_{EB2}}{I_{E1} - I_{E2}}\right|$$

Voltage amplification factor A_V

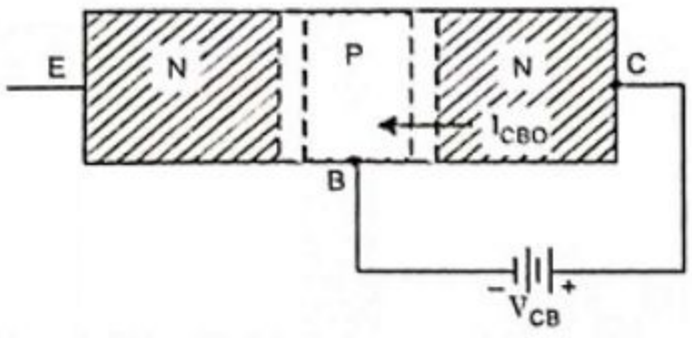
$$A_V = \left.\frac{\Delta V_{CB}}{\Delta V_{EB}}\right| with\ I_E\ const$$

$$A_V = \left|\frac{V_{CB1} - V_{CB2}}{V_{EB1} - V_{EB2}}\right|$$



Note: Both can be determined from the CB input characteristics

Leakage current



(a) Open-Emitter Configuration Showing The Junction Leakage Current, I_{CBO}

- In the dc mode the level of I_C and I_E due to the majority carriers are related by a quantity called alpha

$$\alpha = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E + I_{CBO}$$

- It can then be summarize to $I_C = \alpha I_E$ (ignore I_{CBO} due to small value)
- For ac situations where the point of operation moves on the characteristics curve, an ac alpha defined by

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

- Alpha a common base current gain factor that shows the efficiency by calculating the current percent from current flow from emitter to collector.The value of α is typical from 0.9 ~ 0.998.

- Output resistance r_O

$$r_O = \left.\frac{\Delta V_{CB}}{\Delta I_C}\right| \text{with } I_E \text{ const}$$

- Current amplification factor A_I or α_{ac}

$$\alpha_{ac} = \left.\frac{\Delta I_C}{\Delta I_E}\right| \text{with } V_{CB} \text{ const}$$

Note: Both can be measured from output characteristics

Active region:

- E-B junction forward biased
- C-B junction reverse biased
- I_C is positive, V_{CB} is positive
- I_C increases with I_E

Cut off Region:

- When $I_E = 0$, $I_C = I_{CBO}$
- I_{CBO} is collector to base current with emitter open
- – Below this line we have cut-off region
- – Here both junctions are reverse biased

Saturation Region:

- Region to the left of y-axis (V_{CB} *negative)*
- Here both junctions are forward biased
- I_C decreases exponentially, and eventually changes direction

Common-Emitter Configuration

- It is called common-emitter configuration since :

  - emitter is common or reference to both input and output terminals.

  - emitter is usually the terminal closest to or at ground potential.

- Almost amplifier design is using connection of CE due to the high gain for current and voltage.
- Two set of characteristics are necessary to describe the behavior for CE ;input (base terminal) and output (collector terminal) parameters.

Proper Biasing common-emitter configuration in active region

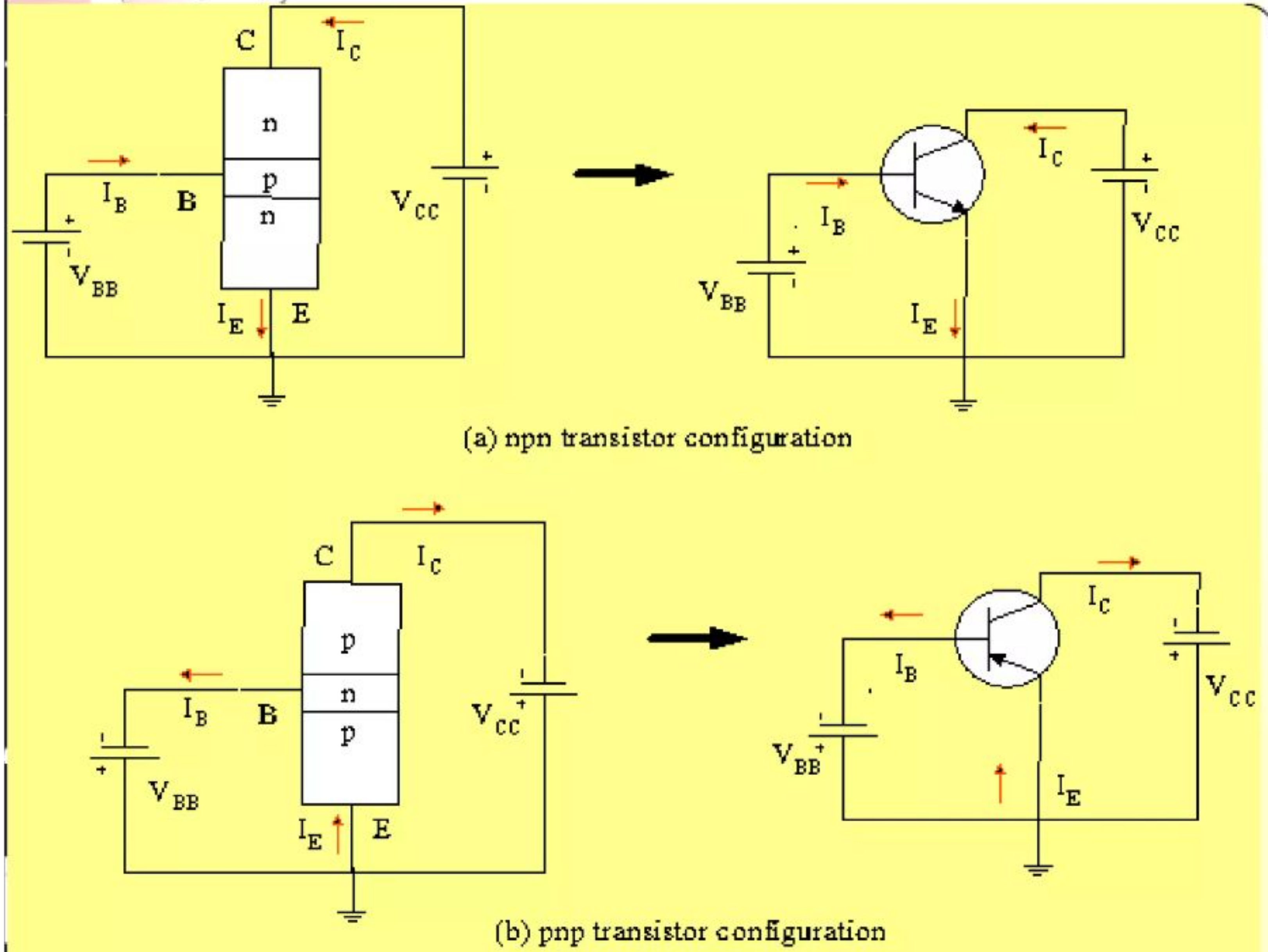
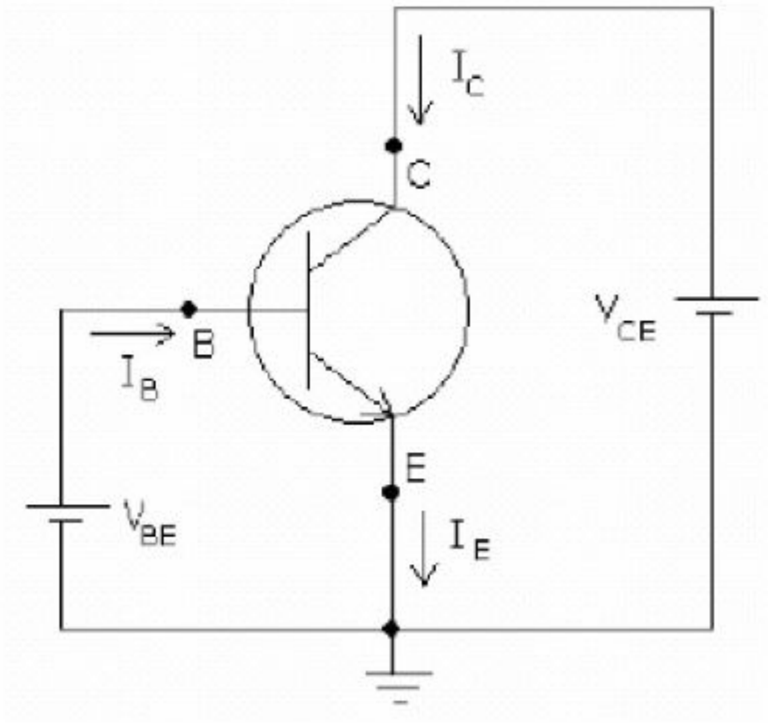


Fig 4.7 : Common-emitter configuration

$$\boxed{I_E = I_C + I_B}$$

Common Emitter configuration



(Resistors are omitted for simplicity)

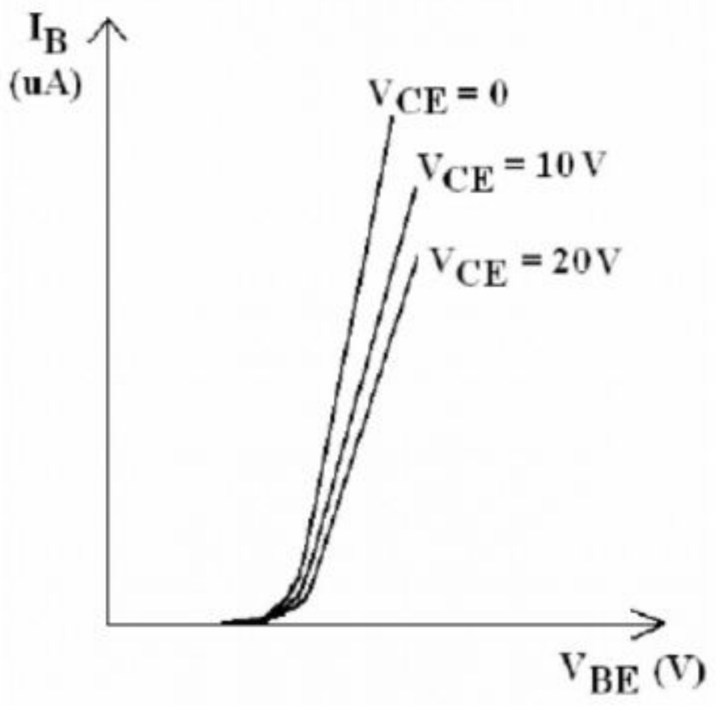
Emitter is common between input and output

– Input voltage: V_{BE} ; *Input current:* I_B

– Output voltage: V_{CE}; *Output current:* I_C

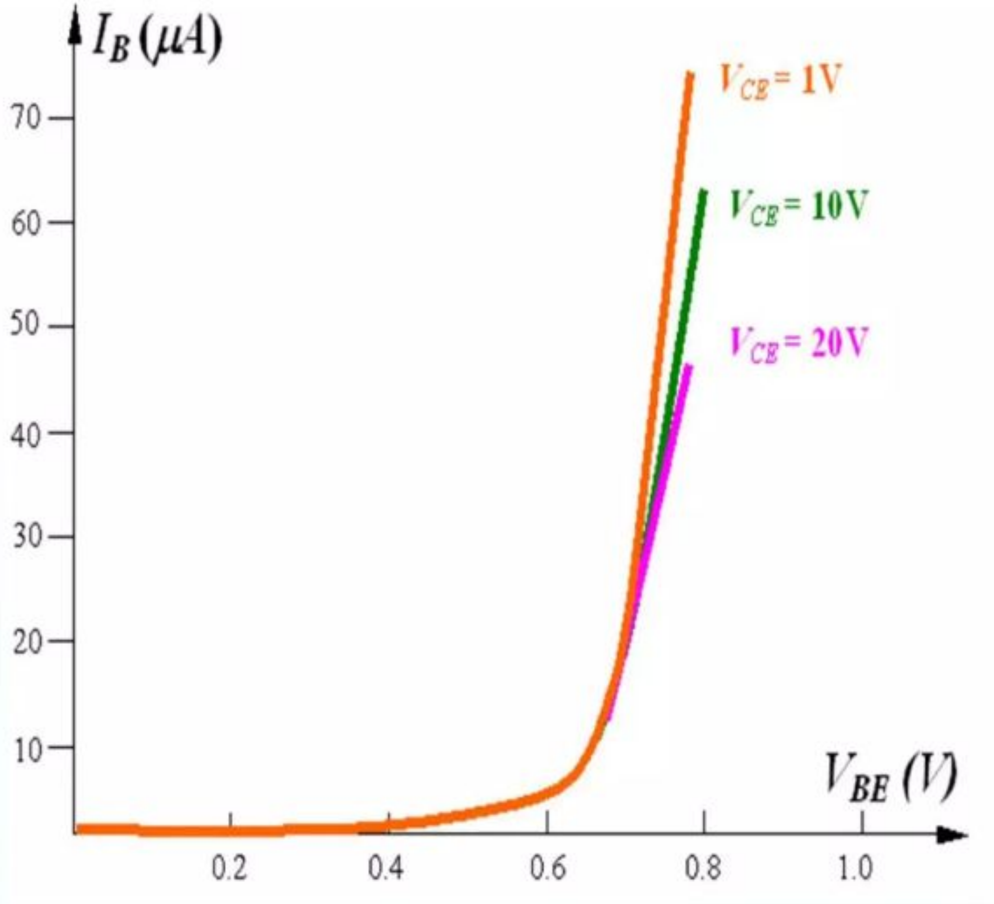
CE input characteristics

- Plot of I_B *versus* V_{BE} *for* various values of V_{CE}
- Similar to diode characteristics
- As V_{CE} *is increased,* I_B decreases only slightly
- Note that second suffix is *E* (for emitter)



Input characteristics

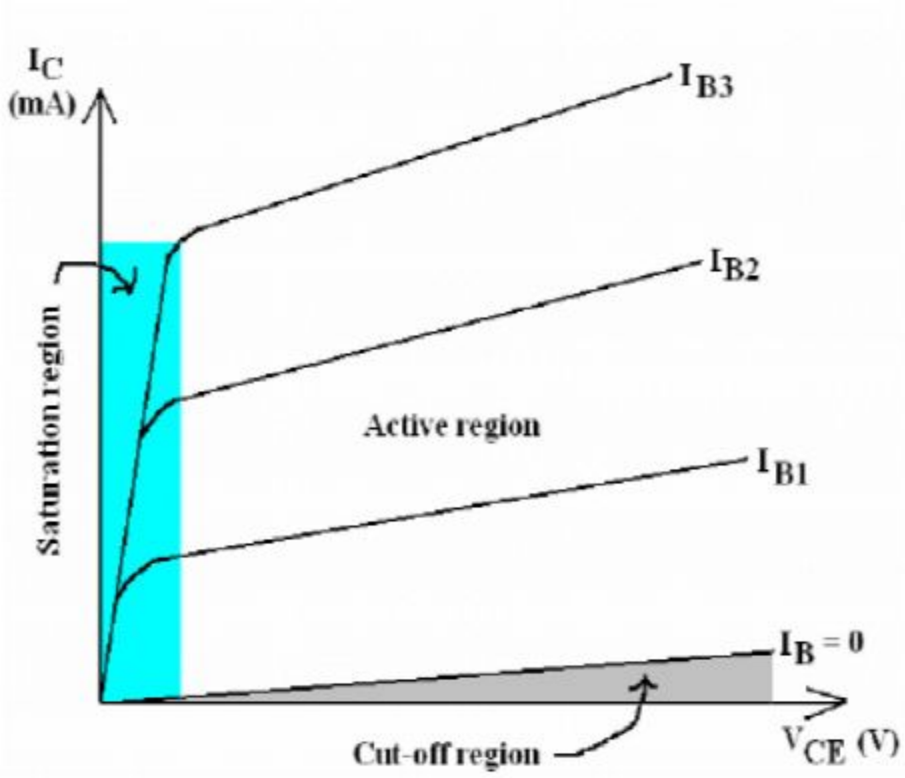
- the input takes the form of a forward-biased *pn* junction
- the input characteristics are therefore similar to those of a semiconductor diode



An input current (I_B) *is a function of an* input voltage (V_{BE}) *for various of output* voltage (V_{CE} *).*

CE output characteristics

– A plot of *I_C versus V_{CE} for various values of I_B*
– Three regions identified:
Active,
Cut-off,
Saturation



Output characteristics

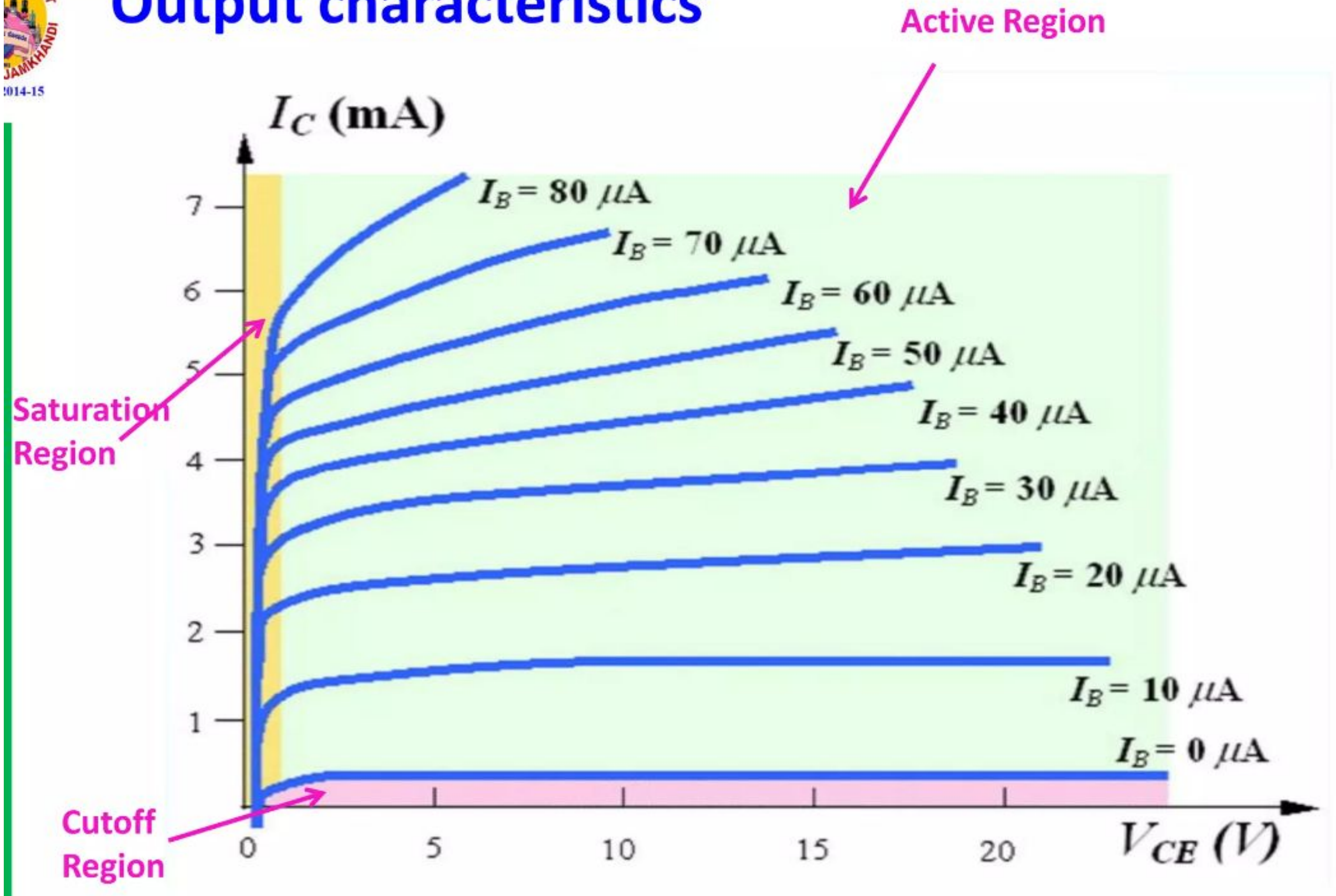


Figure: Output characteristics for common-emitter transistor

- **Output characteristics**
  - The magnitude of I_B *is in μA and not as* horizontal as I_E *in common-base circuit.*
  - The output set relates an output current (I_C) to an output voltage (V_{CE}) *for various of* level of input current (I_B).
- **There are three portions as shown:**

Active region

- The active region, located at upper-right quadrant, has the greatest linearity.
- The curve for *IB are nearly straight and* equally spaced.
- In active region, the *B-E junction is* forward-biased, whereas the *C-B junction* is reverse-biased.
- The active region can be employed for voltage, current or power amplification.

Cutoff region

- The region below *I_B = 0μA is defined as* cutoff region.
- For linear amplification, cutoff region should be avoided.

Saturation region:

- The small portion near the ordinate, is the saturation region, which should be avoided for linear application.
- In the dc mode, the levels of *IC and IB at the* operation point are related by: Normally, β *ranges from 50 to 400.*

$$\beta_{dc} = I_C / I_B$$

For ac situations, β *is defined as*

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}}$$

Common Collector Configuration

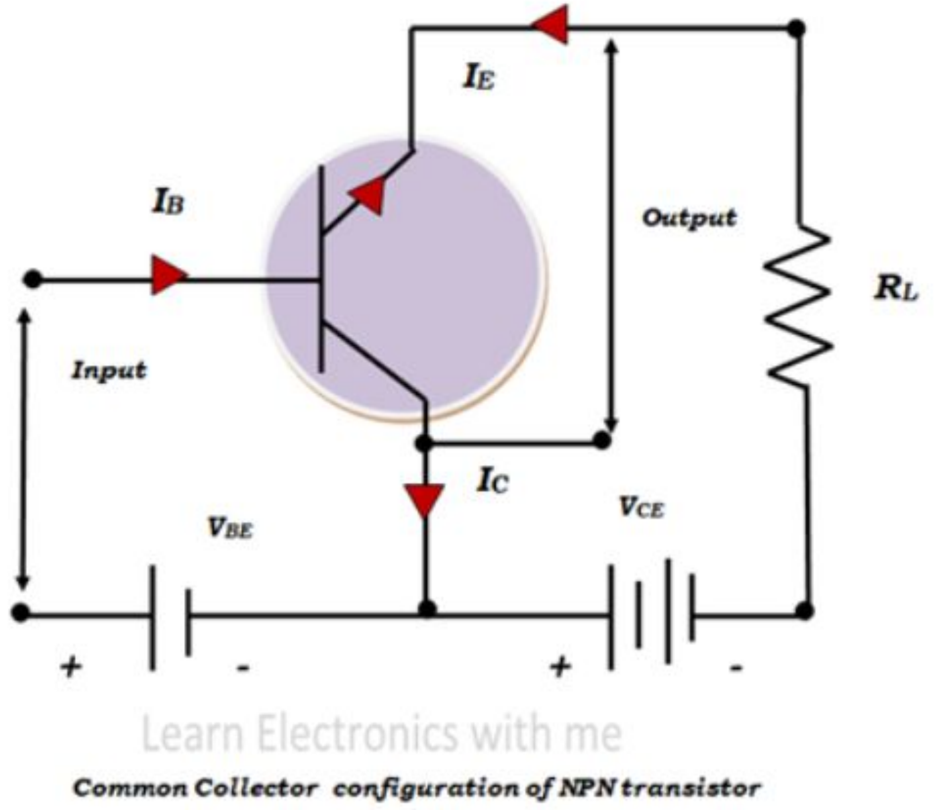
CC :In this circuit the base terminal of the transistor serves as the input, the emitter is the output, and the collector is common to both

Signal input between base and common collector.

Signal output between emitter and common collector.

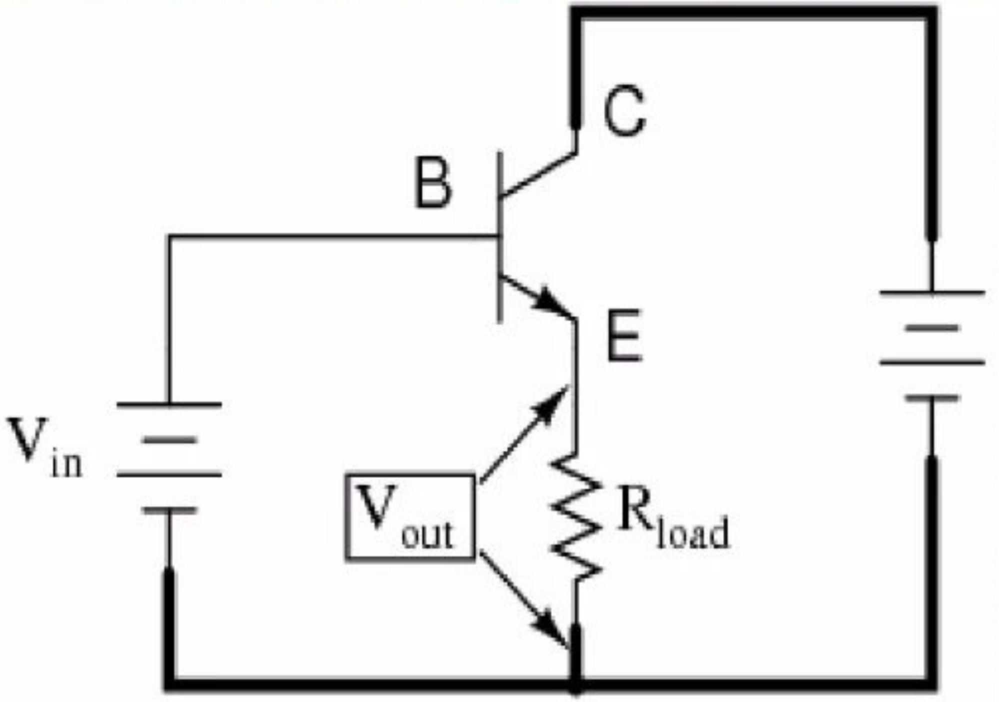
The CC circuit is also known as an emitter follower since the output voltage at the emitter follows the input voltage at the base.

It has voltage gain slightly less than unity.



Common collector configuration of NPN transistor

SIMPLE CIRCUIT DIAGRAM

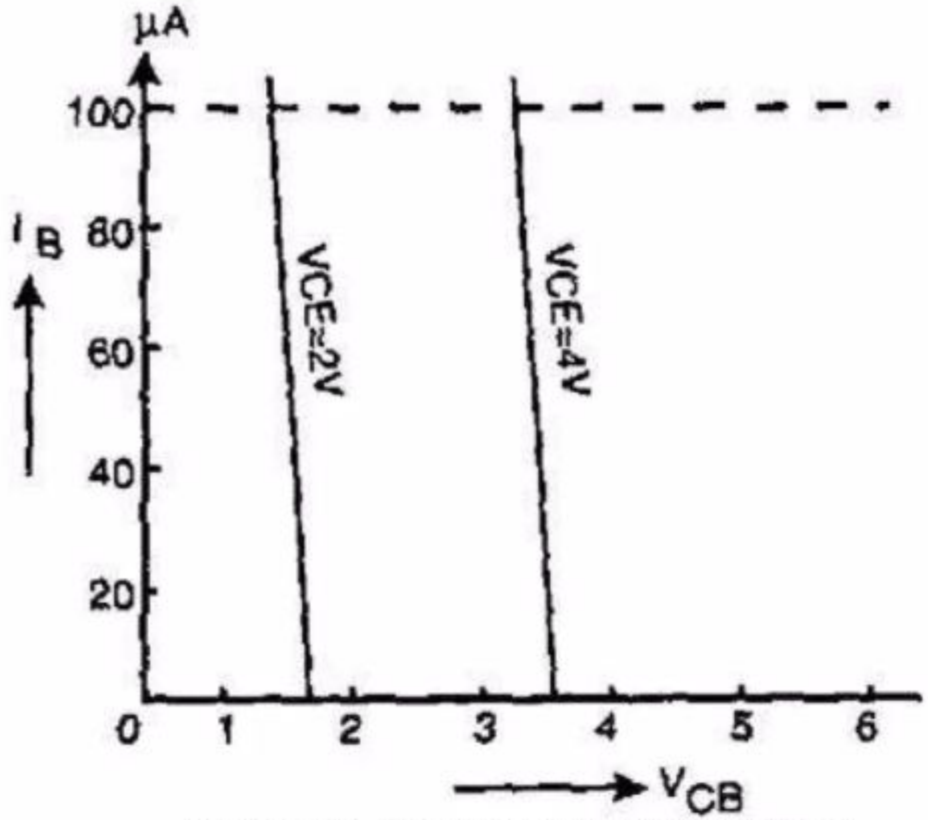


Input is applied to base and collector. Output is from emitter-collector circuit.

WORKING :INPUT CHARACTERISTICS

- The CC input characteristics are quite different from CB & CE characteristics.
- This is because input voltage V_{BC} is largely determined by output voltage V_{EC}
- $V_{EB} = V_{EC} - V_{BC}$

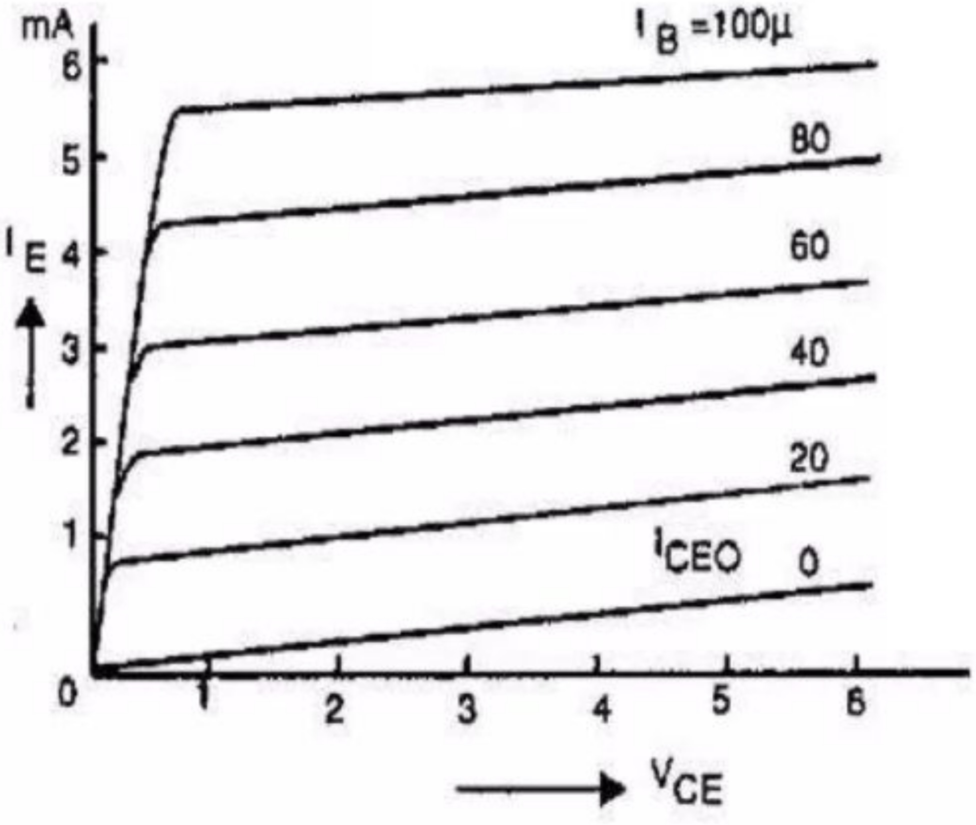
  As V_{BC} increases with V_{EC} constant V_{EB} decreases hence I_B decreases



INPUT CHARACTERISTIC

OUTPUT CHARACTERISTICS

- Here V_{CE} increases I_E also increases.
- Just as in common emitter output characteristics I_C increases with increasing I_B ,so I_E also increases here with increasing I_B
- Hence for constant V_{EC}, I_E increases with I_B



Comparisons of CB,CE & CC Configurations:

Characteristics	CB	CE	CC
Common Terminal for Input and Output	Base Terminal	Emitter Terminal	Collector Terminal
Input voltage applied between	Emitter and Base terminal	Base and Emitter Terminal	Base and Collector Terminal
Output Voltage taken across	Collector and Base Terminal	Collector and Emitter Terminal	Emitter and Collector Terminal
Input Impedance	Very Low(only 50 to 500 ohm)	Medium(500 to 5000 ohm)	Very high(200 to 750 kilo ohm)
Output Impedance	Very High(1 to 10 Mega Ohm)	Medium(50 to 500 kilo ohm)	Very Low(up to 50 ohm)
Input Current	Emitter Current or IE	Base Current or IB	Base Current or IB
Output Current	Collector Current or IC	Collector Current or IC	Emitter Current or IE

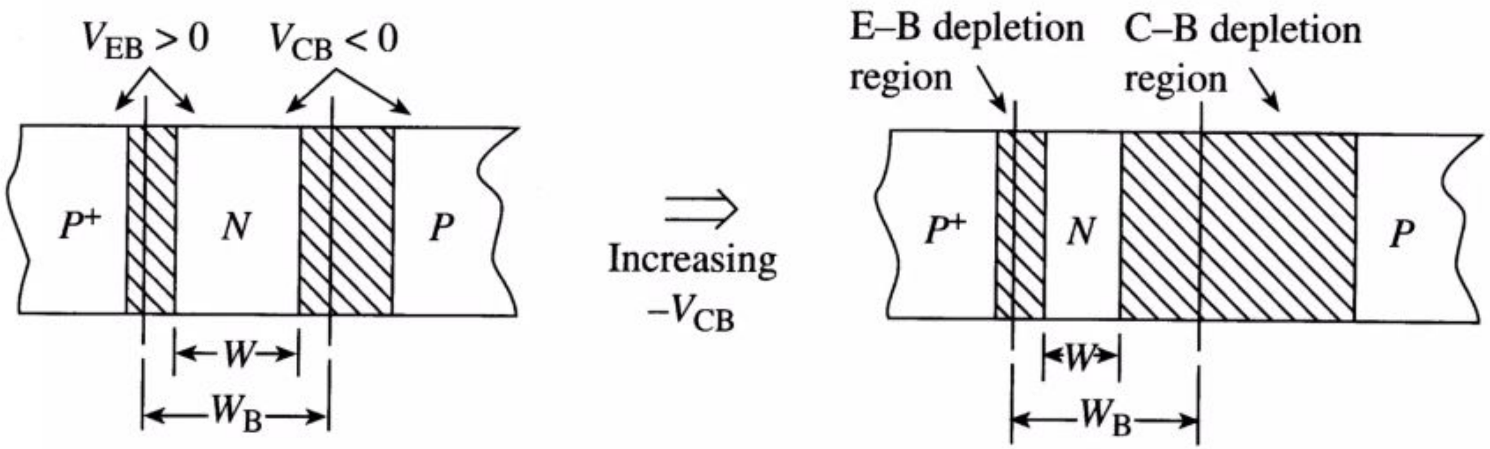
Output Signal Phase	Same phase with input	180 degree out of phase	Same phase with input
Current Gain	Always less than Unity $\alpha = IC/IE$	Between 35 to 500 $\beta = IC/IB$	Very High $\gamma = IE/IB$
Voltage Gain	About 150	About 500	Less Than Unity
Leakage Current	Very Small	Very Large	Very Large
Power Gain	Medium	High	Medium
Application	High Frequency Circuits	RF Signal Processing	Switching Circuits

Base Width Modulation: "Early" Effect

Not in revised syllabus

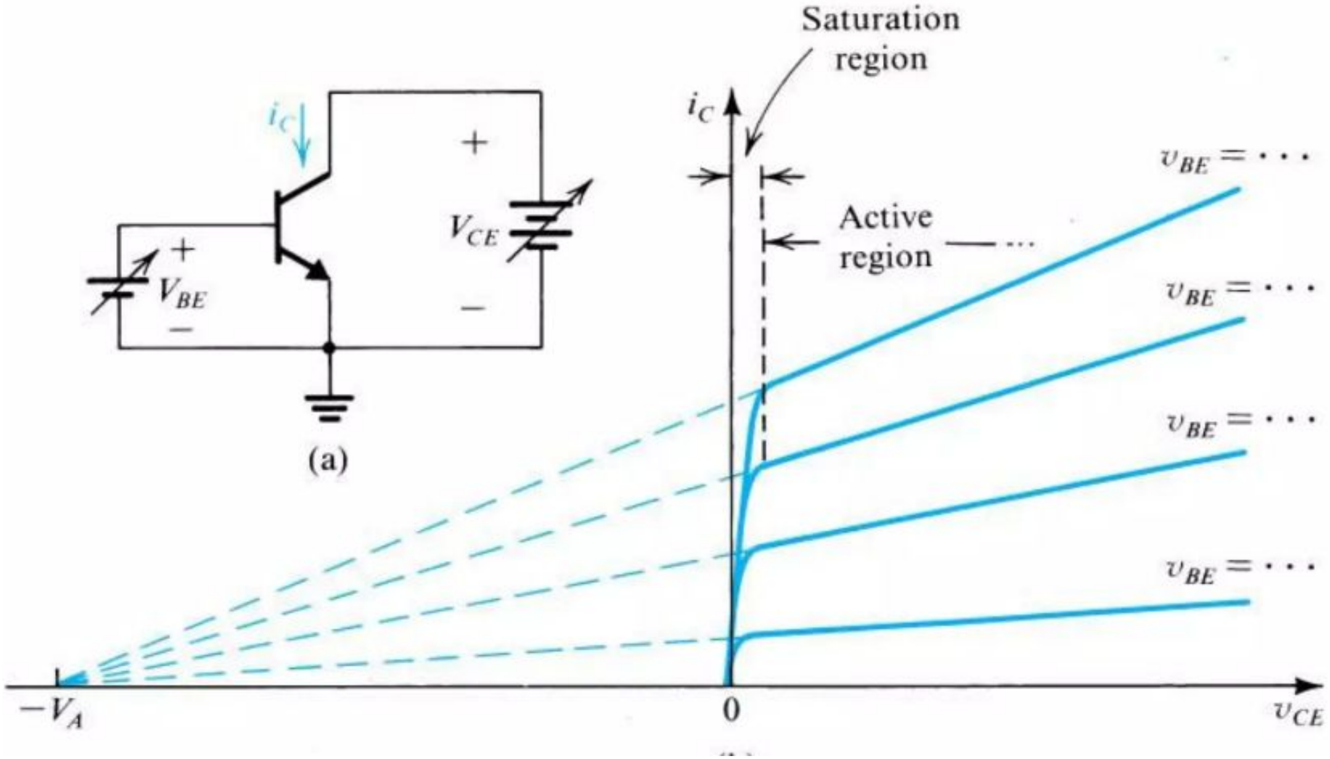
When bias voltages change, depletion widths change and the effective base width will be a function of the bias voltages

Most of the effect comes from the C-B junction since the bias on the collector is usually larger than that on the E-B junction



Base width gets smaller as applied voltages get larger

The Early Effect



Converge ~ at single point called "Early Voltage" (after James Early)

Large "Early Voltage" = Absence of "Base Width Modulation"

Alpha(α),Beta(β) & Gamma(γ)

ALPHA (α): It is a large signal current gain in common bas configuration. It is the ratio of collector current (output current) to the emitter current (input current).

$$\alpha = \frac{Collector\ current}{Emitter\ current}$$

$$\alpha = \frac{I_C}{I_E}$$

Beta (β): It is a current gain factor in the common emitter configuration. It is the ration of collector current (output current) to base current (output current).

$$beta = \frac{I_C}{I_B}$$

Alpha(α),Beta(β) & Gamma(γ)

Gamma (γ): It is a current gain in common collector configuration and It is the ration of emitter current (output current) to base current (input current).

$$\gamma = \frac{I_E}{I_B}$$

It is also called emitter efficiency that how much current is injected from the emitter to base after recombination of minority charge carriers in base. It's value is high compared to α,β.

Relation between α,β and γ in a transistor

α,β and γ are the current gain factors in three CB, CE and CC configurations respectively.

Relation between α, β and γ :

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\gamma = \beta + 1$$

METHODS OF TRANSISTOR BIASING:

There are four types of transistor biasing method,

(i) Base resistor method.

(ii) Emitter bias method.

(iii) Biasing with collector-feedback resistor.

(iv) Voltage-divider bias.

BASE RESISTOR METHOD:

Or, Fixed Biased

In this method, a high resistance R_B (several hundred $k\Omega$) is connected between the base and +ve end of supply for *npn* transistor (See Fig. 1) and between base and negative end of supply for *pnp* transistor. Here, the required zero signal base current is provided by V_{CC} and it flows through R_B. It is because now base is positive w.r.t. emitter i.e. base-emitter junction is forward biased. The required value of zero signal base current I_B (and hence $I_C = \beta I_B$) can be made to flow by selecting the proper value of base resistor R_B.

CIRCUIT ANALYSIS:

- It is required to find the value of R_B so that required collector current flows in the zero signal conditions. Let I_C be the required zero signal collector current.

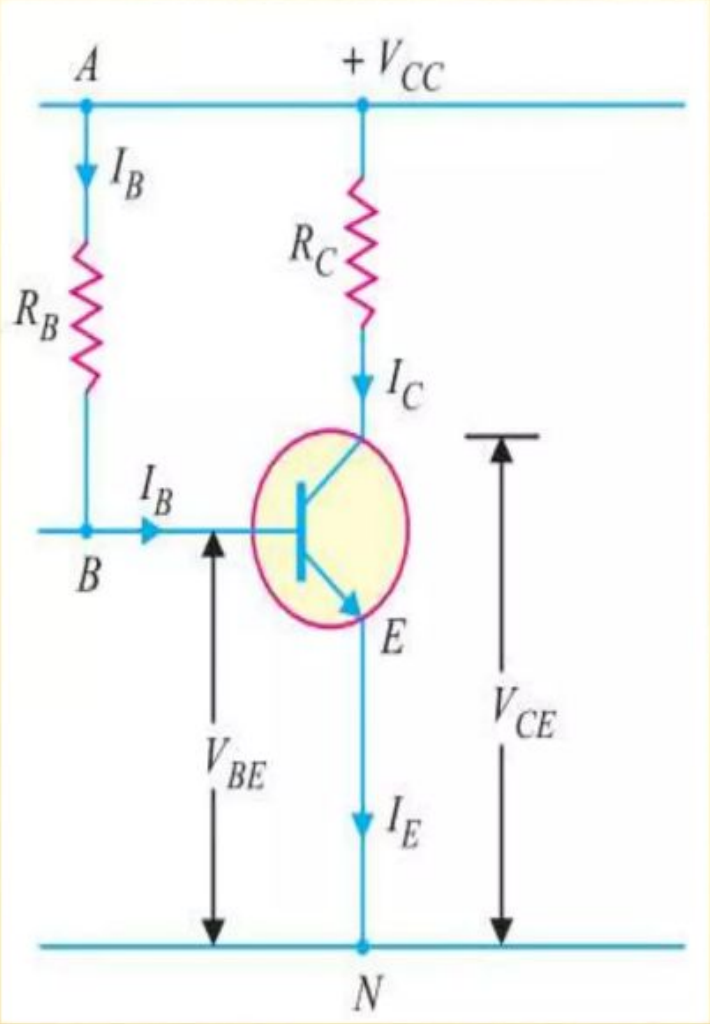
$$\therefore \quad I_B = \frac{I_C}{\beta}$$

Considering the closed circuit *ABENA* and applying Kirchhoff's voltage law, we get,

$$V_{CC} = I_B R_B + V_{BE}$$

$$\text{or} \quad I_B R_B = V_{CC} - V_{BE}$$

$$\therefore \quad R_B = \frac{V_{CC} - V_{BE}}{I_B}$$



CIRCUIT ANALYSIS:

As Vcc and IB are known and VBE can be seen from the transistor manual, therefore, value of RB

can be readily found from exp. (i).

Since VBE is generally quite small as compared to VCC, the former can be neglected with little error. It then follows from exp. (i)

that :

$$R_B = \frac{V_{CC}}{I_B}$$

It may be noted that V_{CC} is a fixed known quantity and I_B is chosen at some suitable value. Hence,

R_B can always be found directly, and for this reason, this method is sometimes called fixed-bias method.

Stability factor. As shown in fig-1

$$\text{Stability factor,} \quad S = \frac{\beta + 1}{1 - \beta\left(\frac{dI_B}{dI_C}\right)}$$

In fixed-bias method of biasing, I_B is independent of I_C so that $dI_B/dI_C = 0$. Putting the value of

$dI_B / dI_C = 0$ in the above expression, we have,

Stability factor, $S = \beta + 1$

Thus the stability factor in a fixed bias is $(\beta + 1)$. This means that I_C changes $(\beta + 1)$ times as

much as any change in I_{CO}. For instance, if $\beta = 100$, then $S = 101$ which means that I_C increases 101

times faster than I_{CO}. Due to the large value of S in a fixed bias, it has poor thermal stability.

ADVANTAGES :

(i) This biasing circuit is very simple as only one resistance RB is required.

(ii) Biasing conditions can easily be set and the calculations are simple.

(iii) There is no loading of the source by the biasing circuit since no resistor is employed across base-emitter junction.

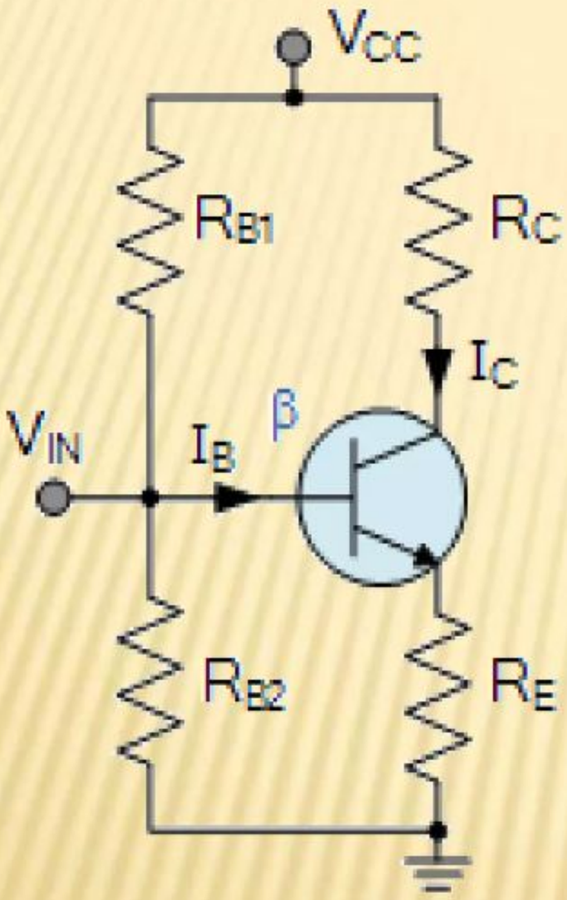
DISADVANTAGES :

(i) This method provides poor stabilization. It is because there is no means to stop a self increase in collector current due to temperature rise and individual variations. For example, if β increases due to transistor replacement, then I_C also increases by the same factor as I_B is constant.

(ii) The stability factor is very high. Therefore, there are strong chances of thermal runaway.

Due to these disadvantages, this method of biasing is rarely employed.

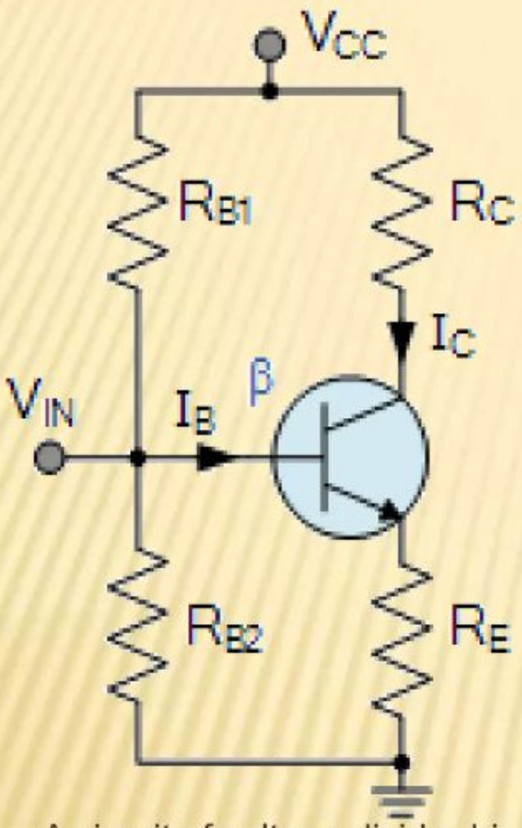
VOLTAGE DIVIDER BIAS METHOD



A circuit of voltage divider bias method

- This is the most widely used method of providing biasing and stabilisation to a transistor. In this method, two resistances R_{B1} and R_{B2} are connected across the supply voltage V_{cc} and providing biasing. The emitter resistance R_E provides stabilisation. The name "voltage divider" comes from the voltage divider formed by R_{B1} and R_{B2}. The voltage drop across R_2 forward biases the base emitter junction. The causes the base current and hence collector current flow in the zero signal conditions.

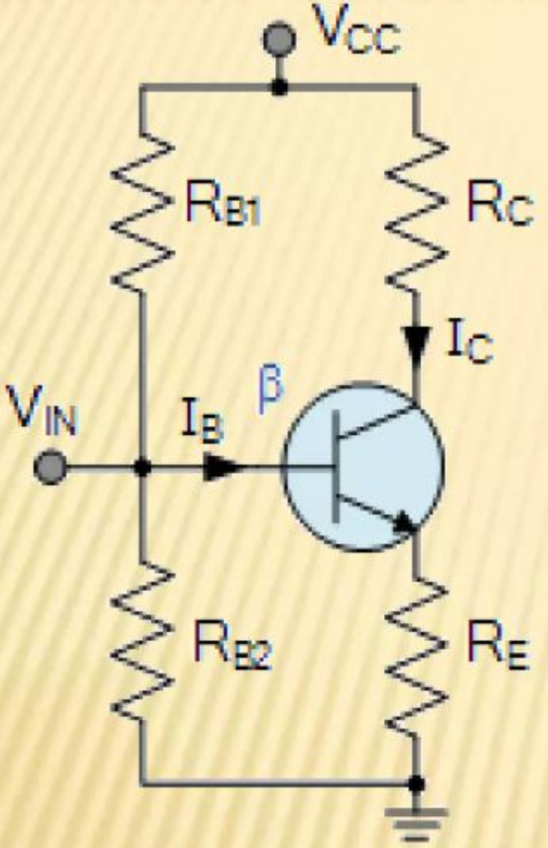
CIRCUIT ANALYSIS



A circuit of voltage divider bias method

- Circuit analysis: Suppose that the current flowing through resistance R_{B1} is I_1. As base current I_B is very small. Therefore, it can be assumed with reasonable accuracy that current flowing through R_{B2} is also I_1.

CIRCUIT ANALYSIS



A circuit of voltage divider bias method

- **Collector Current I_C:**

$$I_1 = V_{CC} \; / R_{B1} + R_{B2}$$

So, voltage across resistance R_{B2} is

$$V_2 = (V_{CC} \, / R_{B1} + R_{B2}) \, R_{B2}$$

Appling KVL to the base circuit of the Fig

$$V_2 = V_{BE} + V_E$$

$$V_2 = V_{BE} + I_E R_E$$

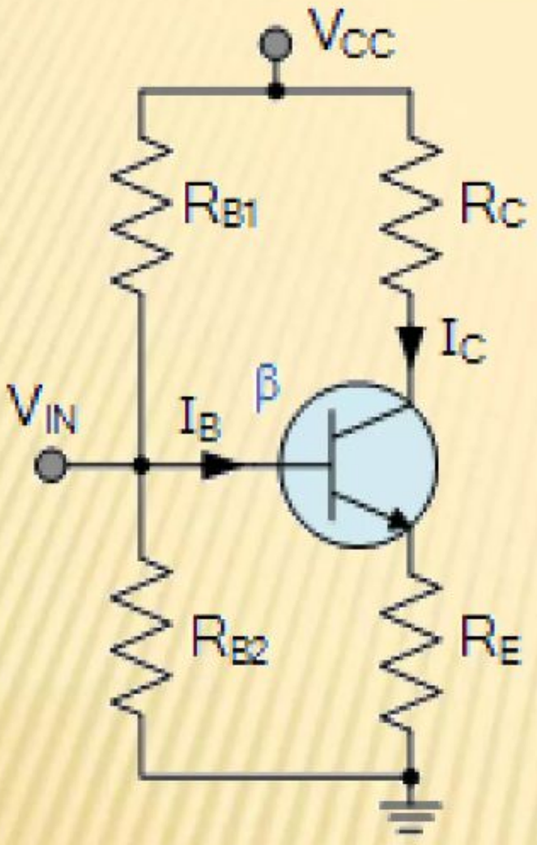
Or, $I_E = (V_2 + V_{BE}) / R_E$

Since, $I_E = I_C$

$$I_C = (V_2 - V_{BE}) / R_E$$

Though I_C depends upon V_{EE} but in practice $V_2 >> V_{BE}$, so that I_C is practically independent of V_{BE}.

CIRCUIT ANALYSIS



A circuit of voltage divider bias method

- **Collector-emitter voltage V_{CE}:** Applying KVL to the collector side.

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E$$
$$= I_C R_C + V_{CE} + I_C R_E$$

[Because $I_E = I_C$]

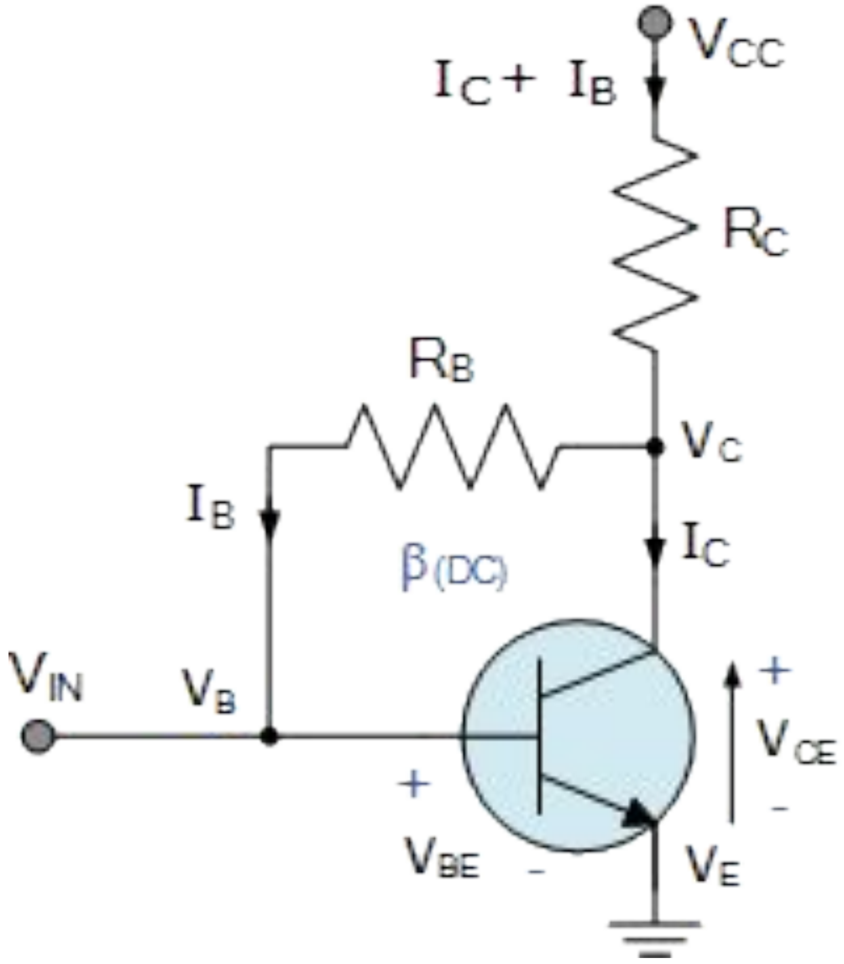
$$= I_C (R_C + R_E) + V_{CE}$$
$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Stabilisation: In this circuit, excellent stabilization is provided by $R_{E.}$ We know that,

$$V_2 = V_{CC} - I_C (R_C + R_E)$$

Suppose the collector current IC increases due to rise in temperature. This will cause the voltage drop across emitter resistance RE to increases. As voltage drop across R2 is independent of IC therefore, VBE decreases. This in turn causes IB to decrease. The reduced value of IB tends to restore IC to the original value.

Collector Feedback Biasing



$$V_C = V_{CC} - R_C(I_C + I_B)$$

$$V_E = 0v$$

$$V_B = V_{BE}$$

$$I_B = \frac{V_C - V_B}{R_B}$$

$$I_C = \beta_{(DC)} I_B$$

$$I_E = (I_C + I_B) \cong I_C$$

Emitter Feedback Configuration

The emitter-feedback bias circuit is designed by adding an emitter resistor to the base-bias circuit. This circuit is used for the **negative feedback**.

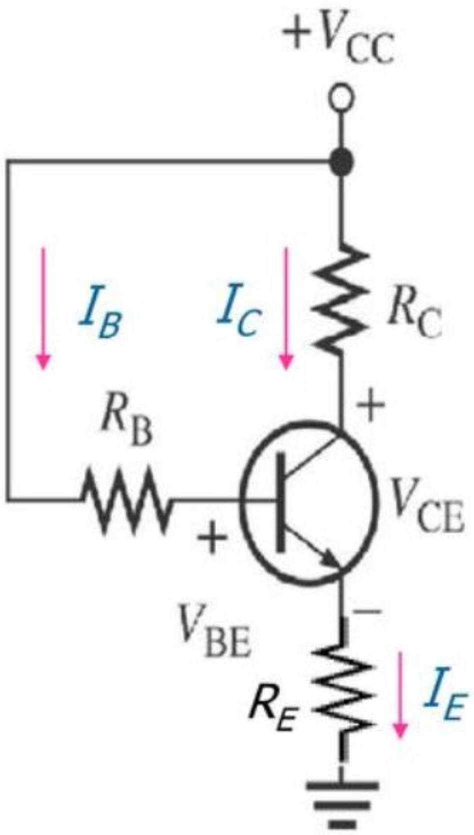
When I_C increases, V_E also increases, causing an increase in V_B because $V_B = V_E + V_{BE}$. This increase in V_B reduces I_B and keeping I_C from increasing.

To determine I_E, KVL is used around the base circuit.

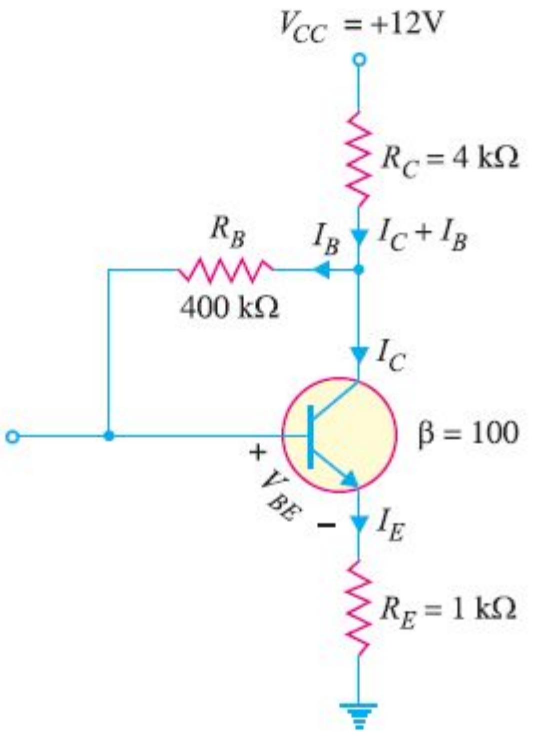
$$-V_{CC} + I_B R_B + V_{BE} + I_E R_E = 0$$

Substituting I_E/β_{DC} for I_B, we obtain

$$I_E = \frac{V_{CC} - V_{BE}}{R_E + \frac{R_B}{\beta_{DC}}}$$



Find the Q-point values (I_C and V_{CE}) for the collector feedback bias circuit shown in Fig. 14.



$$V_{CC} - (I_C + I_B)\,R_C - I_B\,R_B - V_{BE} - I_E\,R_E = 0$$

Now $\quad I_B + I_C \simeq I_C\,;\, I_E \simeq I_C$ and $I_B = \dfrac{I_C}{\beta}$

$\therefore \quad V_{CC} - I_C\,R_C - \dfrac{I_C}{\beta}\,R_B - V_{BE} - I_C\,R_E = 0$

or $\quad I_C\left(R_E + \dfrac{R_B}{\beta} + R_C\right) = V_{CC} - V_{BE}$

$\therefore \quad I_C = \dfrac{V_{CC} - V_{BE}}{R_E + R_B/\beta + R_C}$

Putting the given circuit values, we have,

$$I_C = \frac{12V - 0.7V}{1\ k\Omega + 400\ k\Omega/100 + 4\ k\Omega}$$

$$= \frac{11.3V}{9\ k\Omega} = 1.26\ mA$$

$$V_{CE} = V_{CC} - I_C\,(R_C + R_E)$$

$$= 12V - 1.26\ mA\ (4k\Omega + 1\ k\Omega)$$

$$= 12V - 6.3V = 5.7V$$

$\therefore$ The operating point is **5.7V, 1.26 mA.**

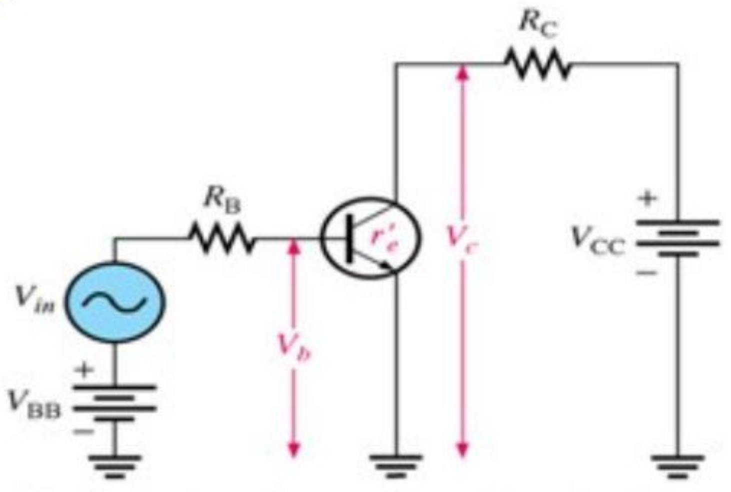
BJT as an Amplifier

- Amplification of a small ac voltage by placing the ac signal source in the base circuit

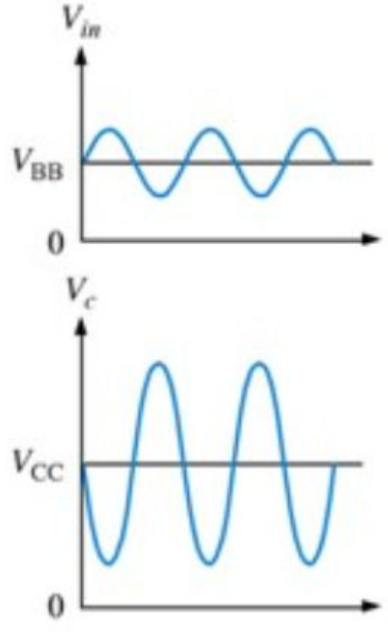
- V_{in} is superimposed on the DC bias voltage V_{BB} by connecting them in series with base resistor R_B:

$$I_C = \beta_{DC} I_B$$

- Small changes in the base current circuit causes large changes in collector current circuit

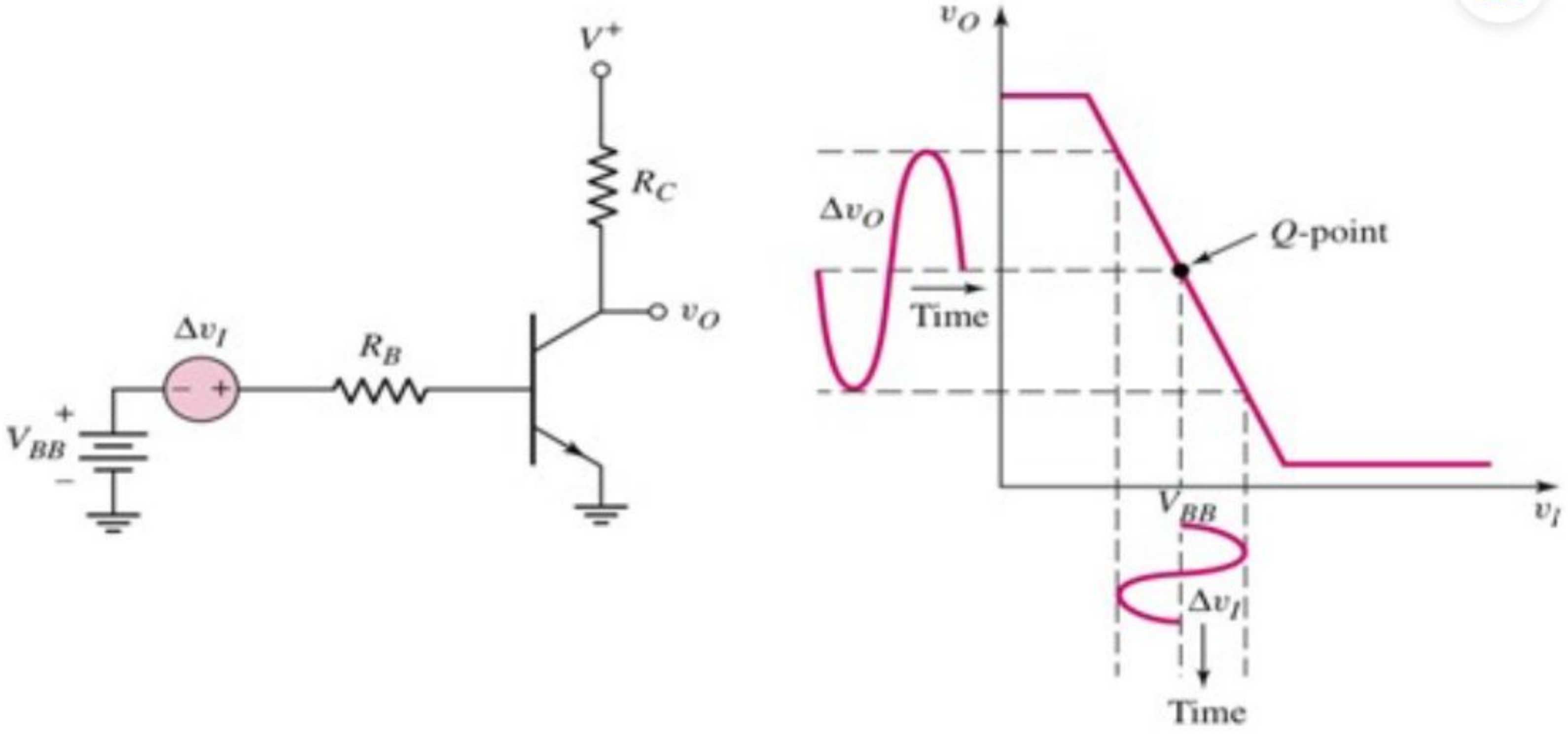


(a) Circuit with ac input voltage V_{in} and dc bias voltage superimposed



(b) Waveforms

BJT as an Amplifier (cont')



(a)

(b)

(a)A bipolar inverter circuit to be used as a time-varying amplifier

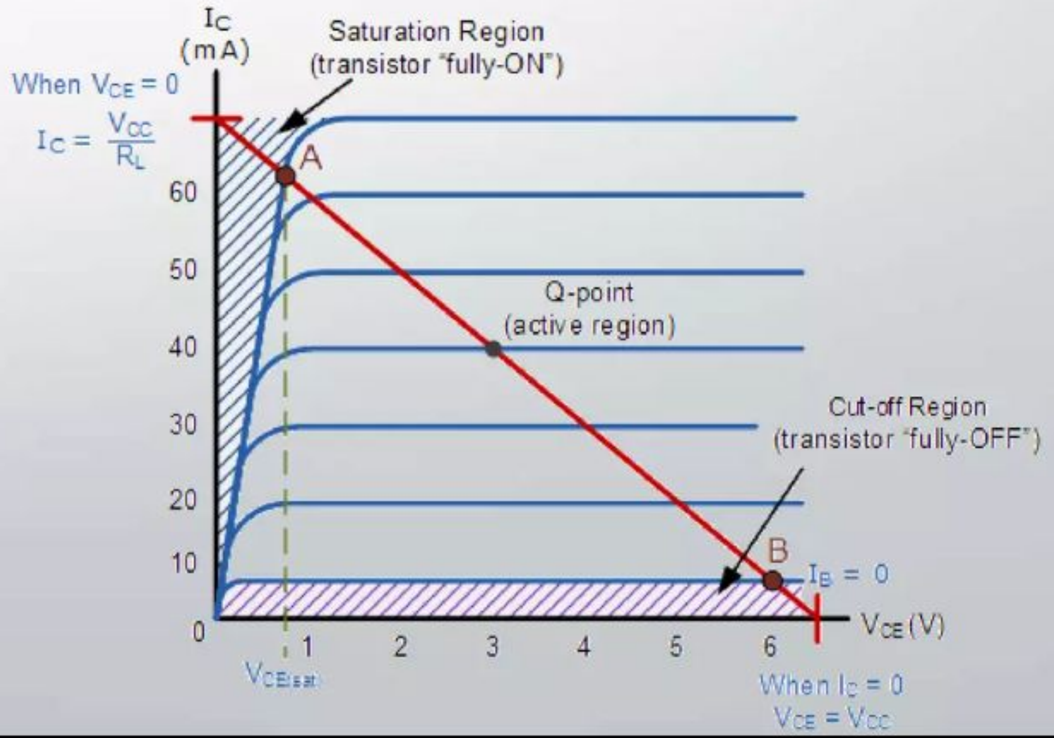
(b) The voltage transfer characteristic

Transistor as a Switch

- When used as an AC signal amplifier, the transistors Base biasing voltage is applied in such a way that it always operates within its "active" region, that is the linear part of the output characteristics curves are used. However, both the NPN & PNP type bipolar transistors can be made to operate as "ON/OFF" type solid state switches by biasing the transistors base differently to that of a signal amplifier.
- Transistor switch operating region
- NPN Transistor and PNP Transistor

  V-I Characteristics curves have seen

  in figure.

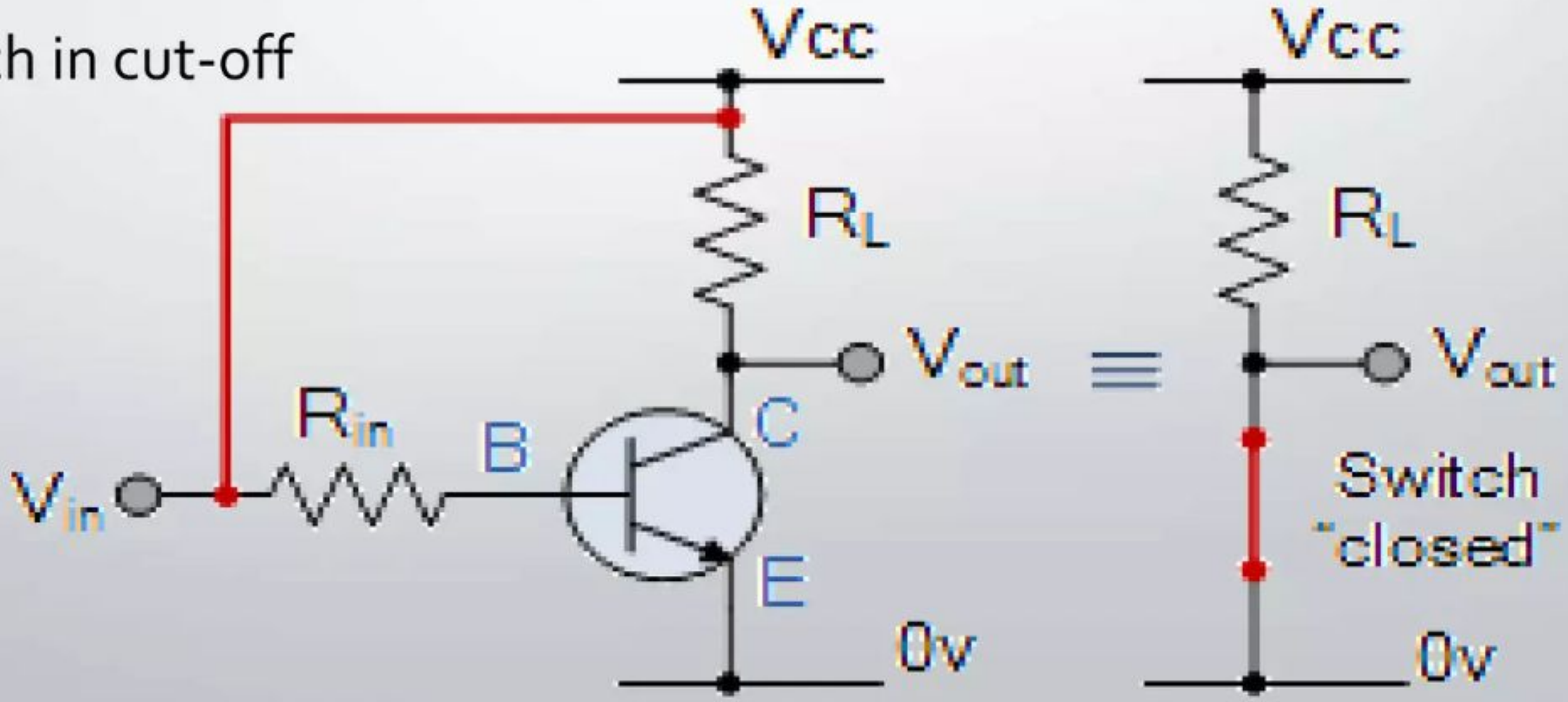


- The areas of operation for a Transistor Switch are known as the Saturation Region and the Cut-off Region. This means then that we can ignore the operating Q-point biasing and voltage divider circuitry required for amplification, and use the transistor as a switch by driving it back and forth between its "fully-OFF" (cut-off) and "fully-ON" (saturation) regions as shown in figure.
- The pink shaded area at the bottom of the curves represents the "Cut-off" region while the blue area to the left represents the "Saturation" region of the transistor. Both these transistor regions are defined as:

1. Cut-off Region

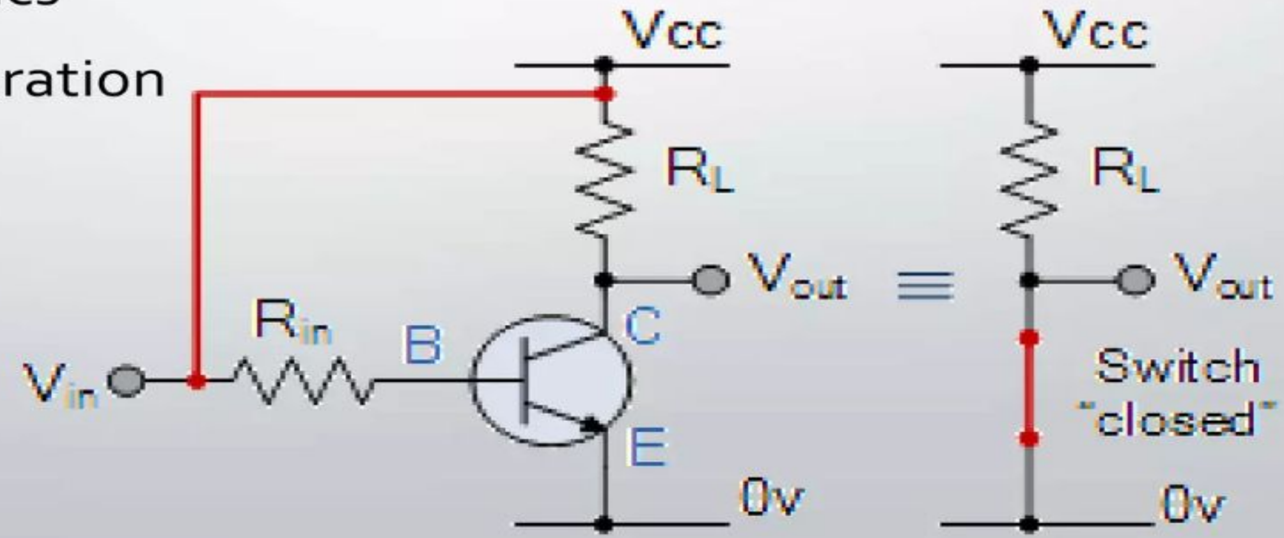
- Then we can define the "cut-off region" or "OFF mode" when using a bipolar transistor as a switch as being, both junctions reverse biased, VB < 0.7v and IC = 0. For a PNP transistor, the Emitter potential must be negative with respect to the Base.

- Here the operating conditions of the transistor are zero input base current (IB), zero output collector current (IC) and maximum collector voltage (VCE) which results in a large depletion layer and no current flowing through the device. Therefore the transistor is switched "Fully-OFF".
- transistor switch in cut-off



2. Saturation Region

- Here the transistor will be biased so that the maximum amount of base current is applied, resulting in maximum collector current resulting in the minimum collector emitter voltage drop which results in the depletion layer being as small as possible and maximum current flowing through the transistor. Therefore the transistor is switched "Fully-ON".
- Saturation Characteristics
- transistor switch in saturation



- Then we can define the "saturation region" or "ON mode" when using a bipolar transistor as a switch as being, both junctions forward biased, VB > 0.7v and IC = Maximum. For a PNP transistor, the Emitter potential must be positive with respect to the Base.

Press Esc to exit full screen

Switching Time

Transistor switching times:

$$t_{on} = t_r + t_d$$

$$t_{off} = t_s + t_f$$

For general purpose transistor, we find:

ts = 120 ns

td = 25 ns

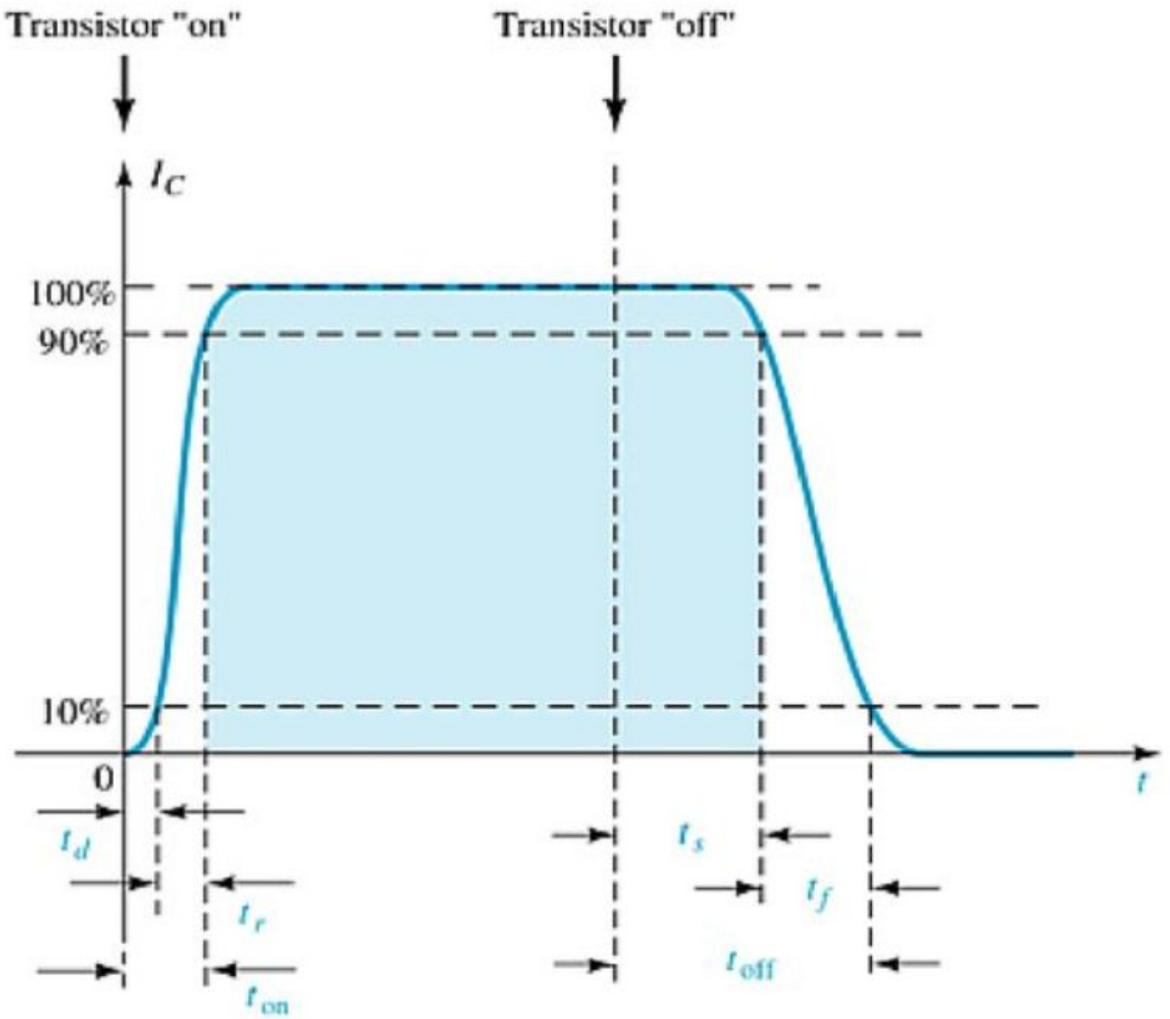
tr = 13 ns

tf = 12 ns

Therefore:

t_{on} = 38 ns

t_{off} = 132 ns



Transistor Switching Time

When the input current is switched off, I_C does not go to zero until after a **turn-off** time t_{off}, made up of a **storage time** (t_s), and a **fall time** (t_f), as illustrated. The fall time is specified as the time required for I_C to go from 90% to 10% of its maximum level. The storage time is the result of charge carriers being trapped in the depletion region when a junction polarity is reversed.

When a transistor is in a saturated on condition, both the collector-base and emitter-base junctions are forward biased. At switch-off both junctions are reverse biased, and before I_C begins to fall, the stored charge carriers must be withdrawn or made to recombine with opposite-type charge carriers.

The Thermal Stability of Operating Point (S_{Ico})

❖ **Stability Factor *S*:-** The stability factor *S, as the change of collector current with respect to the reverse saturation* current, keeping β *and VBE constant. This can be written as:*

The Thermal Stability Factor : S_{Ico}

$$S_{Ico} = \left.\frac{\partial I_c}{\partial I_{co}}\right|_{V_{be},\ \beta}$$

This equation signifies that I_c Changes S_{Ico} times as fast as I_{co}

Differentiating the equation of Collector Current $I_C = (1+\beta)Ico + \beta I_b$ & rearranging the terms we can write

$$S_{Ico} = \frac{1+\beta}{1-\beta\,(\partial I_b/\partial I_C)}$$

It may be noted that Lower is the value of S_{Ico} better is the stability

1. General Expression For Stability Factor S:

In the active region the basic relationship between collector current I_C and base current I_B is given below:

$$I_C = \beta I_B + (1+\beta) I_{CO} \qquad \text{...(12.4)}$$

Differentiating above Eq. (12.4) with respect to I_C considering β to be constant, we have

$$1 = \beta \frac{dI_B}{dI_C} + (1+\beta)\frac{dI_{CO}}{dI_C} = \beta \frac{dI_B}{dI_C} + \frac{1+\beta}{S}$$

So Stability Factor of Transistor,

$$S = \frac{1+\beta}{1-\beta \dfrac{dI_B}{dI_C}} \qquad \text{...(12.5)}$$

The value of dI_B/dI_C depends upon the biasing arrangement used and for determination of the stability factor S, it is only necessary to find the relationship between I_C and I_B.

thermal runaway

- as a transistor heats, its junction temperature increases. This increases the collector current, which forces the junction temperature to increase further, producing more collector current, etc., until the transistor is destroyed.

Another definition of thermal runaway:

as a transistor heats, its junction temperature increases. This increases the collector current, which forces the junction temperature to increase further, producing more collector current, etc., until the transistor is destroyed.

What Is Thermal Runaway ?

The self destruction of unstablized transistor is known as Thermal Runaway.

Common Emitter Configuration:

$$I_C = \beta I_B + (\beta+1) I_{CBO} \text{........................(i)}$$

Collector Current

Base Current

Reverse Saturation Current

I_{CBO} → Minority charge Carriers (M.C)

Temperature (T)